

MongoDB Complete Commands – Student Collection

SECTION 1

1. Database Commands

```
> show dbs
< StudentDB   72.00 KiB
  admin       40.00 KiB
  config      108.00 KiB
  j           16.00 KiB
  local       72.00 KiB
> use studentDB
< switched to db studentDB
> db
< studentDB
> show dbs
< StudentDB   72.00 KiB
  admin       40.00 KiB
  config      100.00 KiB
  j           16.00 KiB
  local       72.00 KiB
studentDB>
```

```
> show collections
< students
> show collections
< students
> show collections
< students
```

```

< students
> db.students.insertMany([
  { rollNo: 101, name: "John", age: 20, department: "CSE", marks: 85, city: "Hyderabad" },
  { rollNo: 102, name: "Alice", age: 21, department: "CSE", marks: 90, city: "Vijayawada" },
  { rollNo: 103, name: "Rahul", age: 20, department: "ECE", marks: 75, city: "Hyderabad" },
  { rollNo: 104, name: "Priya", age: 22, department: "IT", marks: 88, city: "Warangal" },
  { rollNo: 105, name: "Kiran", age: 21, department: "CSE", marks: 95, city: "Hyderabad" },
  { rollNo: 106, name: "Anu", age: 20, department: "IT", marks: 82, city: "Guntur" },
  { rollNo: 107, name: "Ravi", age: 23, department: "ECE", marks: 68, city: "Nellore" },
  { rollNo: 108, name: "Sita", age: 21, department: "CSE", marks: 91, city: "Hyderabad" },
  { rollNo: 109, name: "Arun", age: 22, department: "AI&DS", marks: 87, city: "Kurnool" },
  { rollNo: 110, name: "Meena", age: 20, department: "AI&DS", marks: 93, city: "Hyderabad" }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8e637f2e89e532dbcf6611'),
    '1': ObjectId('6a8e637f2e89e532dbcf6612'),
    '2': ObjectId('6a8e637f2e89e532dbcf6613'),
    '3': ObjectId('6a8e637f2e89e532dbcf6614'),
    '4': ObjectId('6a8e637f2e89e532dbcf6615'),
    '5': ObjectId('6a8e637f2e89e532dbcf6616'),
    '6': ObjectId('6a8e637f2e89e532dbcf6617'),
    '7': ObjectId('6a8e637f2e89e532dbcf6618'),
    '8': ObjectId('6a8e637f2e89e532dbcf6619'),
    '9': ObjectId('6a8e637f2e89e532dbcf661a')
  }
}

```

SECTION 2

```

> db.students.find()
< {
  _id: ObjectId('6a8c8226c5458f40b1b3f3ab'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c8226c5458f40b1b3f3ab'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c8226c5458f40b1b3f3aa'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c8226c5458f40b1b3f3ab'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c8226c5458f40b1b3f3ac'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}

```

```

{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ad'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ae'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3af'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3b0'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3b1'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}

```

```

{
  _id: ObjectId('6a8c028ec5458f40b1b3f3b2')
}
{
  _id: ObjectId('6a8c0349c5458f40b1b3f3b3'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c038ec5458f40b1b3f3b4'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 84
}
{
  _id: ObjectId('6a8c038ec5458f40b1b3f3b5'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf0011'),
  rollNo: 181,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf0012'),
  rollNo: 182,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}

```

SECTION 3

```
{
  _id: ObjectId('6a8e637f2e89e532dbcf6613'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6614'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6615'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6616'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
Type "it" for more
StudentDB>
```

```
> db.students.insertOne({
  rollNo: 111,
  name: "Vijay",
  age: 21,
  department: "CSE",
  marks: 86,
  city: "Hyderabad"
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a8e65302e89e532dbcf661b')
}
StudentDB>
```

```
> db.students.insertMany([
  { rollNo: 112, name: "Pooja", age: 20, department: "IT", marks: 84 },
  { rollNo: 113, name: "Ramesh", age: 22, department: "ECE", marks: 79 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8e65572e89e532dbcf661c'),
    '1': ObjectId('6a8e65572e89e532dbcf661d')
  }
}
```

```
}
> db.students.countDocuments()
< 27
StudentDB >
```

SECTION 4

```
> db.students.findOne()
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3a8'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
```

```
> db.students.findOne({ rollNo: 101 })
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3a8'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
```

```
> db.students.find({ department: "CSE" })
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3a8'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3a9'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ac'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3af'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0349c5458f40b1b3f3b3'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
```

SECTION 5

```
{
  _id: ObjectId('6a8e637f2e89e532dbcf6611'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6612'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6615'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6618'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e65302e89e532dbcf661b'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
```

```
> db.students.find({ name: "John" })
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3a8'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6611'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
StudentDB>
```


SECTION 6

```
> db.students.find({ marks: { $gt: 80 } })
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3ab'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3a9'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ab'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ac'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ad'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
```

```
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3b1'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0349c5458f40b1b3f3b2'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c038ec5458f40b1b3f3b4'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 84
}
{
  _id: ObjectId('6a8e637f2e89e532dbc6611'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbc6612'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8e637f2e89e532dbc6614'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
```

SECTION 7

```
< 27
> db.students.find({ marks: { $gt: 80 } })
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3a8'),
  rollNo: 181,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3a9'),
  rollNo: 182,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ab'),
  rollNo: 184,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ac'),
  rollNo: 185,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ad'),
  rollNo: 186,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3af'),
  rollNo: 188,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
```

```
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6616'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6618'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6619'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf661a'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e65302e89e532dbcf661b'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e65572e89e532dbcf661c'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 84
}
StudentDB >
```

```
> db.students.find(( marks: { $lt: 80 } ))
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3aa'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ae'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8c038ec5458f40b1b3f3b5'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
{
  _id: ObjectId('6a8e637f2e89e532dbc6613'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbc6617'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8e65572e89e532dbc661d'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
}
```

```

    }
    > db.students.find({ marks: { $lte: 80 } })
    < {
      _id: ObjectId('6a8c8226c5458f40b1b3f3aa'),
      rollNo: 103,
      name: 'Rahul',
      age: 20,
      department: 'ECE',
      marks: 75,
      city: 'Hyderabad'
    }
    {
      _id: ObjectId('6a8c8226c5458f40b1b3f3ae'),
      rollNo: 107,
      name: 'Ravi',
      age: 23,
      department: 'ECE',
      marks: 68,
      city: 'Nellore'
    }
    {
      _id: ObjectId('6a8c838ec5458f40b1b3f3b5'),
      rollNo: 113,
      name: 'Ramesh',
      age: 22,
      department: 'ECE',
      marks: 79
    }
    {
      _id: ObjectId('6a8e637f2e89e532dbcf6613'),
      rollNo: 103,
      name: 'Rahul',
      age: 20,
      department: 'ECE',
      marks: 75,
      city: 'Hyderabad'
    }
    {
      _id: ObjectId('6a8e637f2e89e532dbcf6617'),
      rollNo: 107,
      name: 'Ravi',
      age: 23,
      department: 'ECE',
      marks: 68,
      city: 'Nellore'
    }
    {
      _id: ObjectId('6a8e65572e89e532dbcf661d'),
      rollNo: 113,
      name: 'Ramesh',
      age: 22,
      department: 'ECE',
      marks: 79
    }
  }
}
```

```
> db.students.find({ department: { $ne: "CSE" } })
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3aa'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ab'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ad'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ae'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3b0'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3b1'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
```

```

> db.students.find({ marks: { $eq: 90 } })
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3a9'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8e637f2e89e532dbc6612'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
StudentDB >

```

```

> db.students.find({ marks: { $gt: 80 } })
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3a8'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3a9'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ab'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ac'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ad'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}

```



```
> db.students.find({ marks: { $lt: 80 } })
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3aa'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ae'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8c038ec5458f40b1b3f3b5'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6613'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6617'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8e65572e89e532dbcf661d'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
}
StudentDB>
```

```

> db.students.find({ marks: { $lte: 80 } })
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3aa'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ae'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8c038ec5458f40b1b3f3b5'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6613'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6617'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8e65572e89e532dbcf661d'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}

```

```

}
> db.students.find({ marks: { $lte: 80 } })
< {
  _id: ObjectId('6a8c0226c5458f40b1b3f3aa'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c0226c5458f40b1b3f3ae'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8c038ec5458f40b1b3f3b5'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6613'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8e637f2e89e532dbcf6617'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8e65572e89e532dbcf661d'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}

```

```

db.students.find({
  $and: [
    { department: "CSE" },
    { marks: { $gt: 85 } }
  ]
})
[
  {
    _id: ObjectId('6a8c0226c5458f40b1b3f3a9'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  }
  {
    _id: ObjectId('6a8c0226c5458f40b1b3f3ac'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a8c0226c5458f40b1b3f3af'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a8c0349c5458f40b1b3f3b3'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a8e637f2e89e532dbcf6612'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  }
  {
    _id: ObjectId('6a8e637f2e89e532dbcf6615'),

```

```

> db.students.find({
  $and: [
    { department: "CSE" },
    { marks: { $gt: 85 } }
  ]
})
[
  {
    _id: ObjectId('6a8c0226c5458f40b1b3f3a9'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  }
  {
    _id: ObjectId('6a8c0226c5458f40b1b3f3ac'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a8c0226c5458f40b1b3f3af'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a8c0349c5458f40b1b3f3b3'),
    rollNo: 111,
    name: 'Vijay',
    age: 21,
    department: 'CSE',
    marks: 86,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a8e637f2e89e532dbcf6612'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  }
  {
    _id: ObjectId('6a8e637f2e89e532dbcf6615'),

```

```

}
db.students.find({
  marks: { $not: { $gt: 80 } }
})
[
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3aa'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3ae'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  },
  {
    _id: ObjectId('6a8c828ac5458f40b1b3f3b2')
  },
  {
    _id: ObjectId('6a8c838ec5458f40b1b3f3b5'),
    rollNo: 113,
    name: 'Ramesh',
    age: 22,
    department: 'ECE',
    marks: 79
  },
  {
    _id: ObjectId('6a8e637f2e89e532dbcf6613'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a8e637f2e89e532dbcf6617'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  },
  {
    _id: ObjectId('6a8e65572e89e532dbcf661d'),
    rollNo: 113,
    name: 'Ramesh',

```

```

}
> db.students.find({
  $nor: [
    { department: "CSE" },
    { marks: { $gt: 90 } }
  ]
})
[
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3aa'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  },
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3ab'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  },
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3ad'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  },
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3ae'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  },
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3b0'),
    rollNo: 100,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    city: 'Kurnool'
  },
  {
    _id: ObjectId('6a8c828ac5458f40b1b3f3b2')
  },

```

```

db.students.find({
  department: { $in: ["CSE", "IT"] }
})
[
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3a8'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 85,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3a9'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    city: 'Vijayawada'
  }
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3ab'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    city: 'Warangal'
  }
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3ac'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3ad'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  }
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3af'),
    rollNo: 108,
    name: 'Sita',
    age: 21,
    department: 'CSE',

```

```

> db.students.find({
  department: { $nin: ["CSE", "IT"] }
})
< [
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3aa'),
    rollNo: 103,
    name: 'Rahul',
    age: 20,
    department: 'ECE',
    marks: 75,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3ae'),
    rollNo: 107,
    name: 'Ravi',
    age: 23,
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  }
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3b0'),
    rollNo: 109,
    name: 'Arun',
    age: 22,
    department: 'AI&DS',
    marks: 87,
    city: 'Kurnool'
  }
  {
    _id: ObjectId('6a8c8226c5458f40b1b3f3b1'),
    rollNo: 110,
    name: 'Meena',
    age: 20,
    department: 'AI&DS',
    marks: 93,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a8c828ac5458f40b1b3f3b2')
  }
  {
    _id: ObjectId('6a8c838ec5458f40b1b3f3b5'),
    rollNo: 113,
    name: 'Ramesh',
    age: 22,
    department: 'ECE',
    marks: 79
  }
  {
    _id: ObjectId('6a8e637f2e89e532dbc6613'),
    rollNo: 103,
    name: 'Rahul',

```

```

]
db.students.updateOne(
  { rollNo: 101 },
  { $set: { marks: 92 } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
db.students.updateMany(
  { department: "CSE" },
  { $set: { status: "Active" } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 10,
  modifiedCount: 10,
  upsertedCount: 0
}
db.students.updateOne(
  { rollNo: 101 },
  { $inc: { marks: 5 } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
db.students.updateOne(
  { rollNo: 101 },
  { $inc: { marks: -2 } }
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB>

```

```

> db.students.updateMany(
  {},
  { $rename: { "city": "location" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 27,
  modifiedCount: 22,
  upsertedCount: 0
}
StudentDB>

```

```

> db.students.updateOne(
  { rollNo: 101 },
  { $set: { attendance: 90 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
StudentDB>

```

```

< switched to db StudentDB
> db.students.deleteOne({ rollNo: 103 })
< {
  acknowledged: true,
  deletedCount: 1
}
StudentDB>

```

```

> db.students.findOne({ rollNo: 103 })
< {
  _id: ObjectId('6a8e637f2e89e532dbcf6613'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  location: 'Hyderabad'
}
StudentDB>

```

```

> db.students.deleteMany({ marks: { $lt: 40 } })
< {
  acknowledged: true,
  deletedCount: 0
}
StudentDB>

```

```

> db.students.deleteMany({})
< {
  acknowledged: true,
  deletedCount: 26
}
StudentDB>

```

```

> db.students.deleteMany({})
< {
  acknowledged: true,
  deletedCount: 26
}
> db.students.find().sort({ marks: 1 })
<
> db.students.find().sort({ marks: -1 })
<
StudentDB>

```

```
> db.students.deleteMany({})
< {
  acknowledged: true,
  deletedCount: 26
}
> db.students.find().sort({ marks: 1 })
<
> db.students.find().sort({ marks: -1 })
<
> db.students.find().sort({ department: 1, marks: -1 })
<
> db.students.find().limit(5)
<
> db.students.find().sort({ marks: -1 }).limit(3)
<
> db.students.find().skip(5)
<
> db.students.find().skip(2).limit(3)
<
StudentDB>
```

```

> db.students.find(
  {},
  { _id: 0, name: 1, marks: 1 }
)
< {
  name: 'John',
  marks: 95
}
{
  name: 'Alice',
  marks: 90
}
{
  name: 'Rahul',
  marks: 75
}
{
  name: 'Priya',
  marks: 88
}
{
  name: 'Kiran',
  marks: 95
}
{
  name: 'Anu',
  marks: 82
}
{
  name: 'Ravi',
  marks: 68
}
{
  name: 'Sita',
  marks: 91
}
{
  name: 'Arun',
  marks: 87
}
{
  name: 'Heena',
  marks: 93
}
{
  name: 'Vijay',
  marks: 86
}
{
  name: 'Pooja',
  marks: 84
}
}

```

```

> db.students.find(
  {},
  { _id: 0, name: 1, department: 1, marks: 1 }
)
< {
  name: 'John',
  department: 'CSE',
  marks: 95
}
{
  name: 'Alice',
  department: 'CSE',
  marks: 90
}
{
  name: 'Rahul',
  department: 'ECE',
  marks: 75
}
{
  name: 'Priya',
  department: 'IT',
  marks: 88
}
{
  name: 'Kiran',
  department: 'CSE',
  marks: 95
}
{
  name: 'Anu',
  department: 'IT',
  marks: 82
}
{
  name: 'Ravi',
  department: 'ECE',
  marks: 68
}
{
  name: 'Sita',
  department: 'CSE',
  marks: 91
}
{
  name: 'Arun',
  department: 'AI&DS',
  marks: 87
}
{
  name: 'Heena',
  department: 'AI&DS',
  marks: 93
}
}

```

```

> db.students.find(
  {},
  { age: 0 }
)
< {
  _id: ObjectId('6a9538c497af2a8638892557'),
  rollNo: 101,
  name: 'John',
  department: 'CSE',
  marks: 95,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a8638892558'),
  rollNo: 102,
  name: 'Alice',
  department: 'CSE',
  marks: 90,
  status: 'Active',
  location: 'Vijayawada'
}
{
  _id: ObjectId('6a9538c497af2a8638892559'),
  rollNo: 103,
  name: 'Rahul',
  department: 'ECE',
  marks: 75,
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a863889255a'),
  rollNo: 104,
  name: 'Priya',
  department: 'IT',
  marks: 88,
  location: 'Warangal'
}
{
  _id: ObjectId('6a9538c497af2a863889255b'),
  rollNo: 105,
  name: 'Kiran',
  department: 'CSE',
  marks: 95,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a863889255c'),
  rollNo: 106,
  name: 'Anu',
  department: 'IT',
  marks: 82,
  location: 'Guntur'
}
}

```


SECTION 14

```
> db.students.distinct("department")
< [ 'AI&DS', 'CSE', 'ECE', 'IT' ]
StudentDB>
```

```
> db.students.distinct("city")
< []
StudentDB>
```

```
> db.students.updateMany(
  {},
  { $rename: { "city": "location" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 13,
  modifiedCount: 0,
  upsertedCount: 0
}
StudentDB>
```

```
> db.students.distinct("location")
< [
  'Guntur',
  'Hyderabad',
  'Kurnool',
  'Nellore',
  'Vijayawada',
  'Warangal'
]
StudentDB>
```

```
> db.students.distinct(
  "department",
  { marks: { $gte: 80 } }
)
< [ 'AI&DS', 'CSE', 'IT' ]
StudentDB>
```

SECTION 15

```
> db.students.createIndex({ rollNo: 1 })
< rollNo_1
> db.students.createIndex({ rollNo: 1 })
< rollNo_1
StudentDB>
```

```
> db.students.createIndex({ rollNo: 1 })
< rollNo_1
> db.students.createIndex({ name: 1 })
< name_1
> db.students.createIndex({ department: 1, marks: -1 })
< department_1_marks_-1
StudentDB>
```

```
> db.students.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { rollNo: 1 }, name: 'rollNo_1' },
  { v: 2, key: { name: 1 }, name: 'name_1' },
  {
    v: 2,
    key: { department: 1, marks: -1 },
    name: 'department_1_marks_-1'
  }
]
StudentDB>
```

```
> db.students.dropIndex("name_1")
< { nIndexesWas: 4, ok: 1 }
StudentDB>
```

```
> db.students.dropIndexes()
< {
  nIndexesWas: 3,
  msg: 'non-_id indexes dropped for collection',
  ok: 1
}
StudentDB>
```

SECTION 16

```

> db.students.aggregate([
  { $match: {} }
])
< {
  _id: ObjectId('6a9538c497af2a8638092557'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 95,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a8638092558'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  status: 'Active',
  location: 'Vijayawada'
}
{
  _id: ObjectId('6a9538c497af2a8638092559'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a863809255a'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  location: 'Warangal'
}
{
  _id: ObjectId('6a9538c497af2a863809255b'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  status: 'Active',
  location: 'Hyderabad'
}

```

```

{
  _id: ObjectId('6a9538c497af2a863809255e'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a863809255f'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  location: 'Kurnool'
}
{
  _id: ObjectId('6a9538c497af2a8638092560'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a8638092561'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a8638092562'),
  rollNo: 112,
  name: 'Pooja',
  age: 20,
  department: 'IT',
  marks: 84
}
{
  _id: ObjectId('6a9538c497af2a8638092563'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}

```

StudentDB>

```

db.students.aggregate([
  { $match: { marks: { $gt: 80 } } }
])
[
  {
    _id: ObjectId('6a9538c497af2a8638092557'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  }
  {
    _id: ObjectId('6a9538c497af2a8638092558'),
    rollNo: 102,
    name: 'Alice',
    age: 21,
    department: 'CSE',
    marks: 90,
    status: 'Active',
    location: 'Vijayawada'
  }
  {
    _id: ObjectId('6a9538c497af2a863809255a'),
    rollNo: 104,
    name: 'Priya',
    age: 22,
    department: 'IT',
    marks: 88,
    location: 'Warangal'
  }
  {
    _id: ObjectId('6a9538c497af2a863809255b'),
    rollNo: 105,
    name: 'Kiran',
    age: 21,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  }
  {
    _id: ObjectId('6a9538c497af2a863809255c'),
    rollNo: 106,
    name: 'Anu',
    age: 20,
    department: 'IT',
    marks: 82,
    location: 'Guntur'
  }
  {
    _id: ObjectId('6a9538c497af2a863809255e'),
    rollNo: 108,
    name: 'Sita',

```

```

db.students.aggregate([
  {
    $group: {
      _id: null,
      lowestMarks: { $min: "$marks" }
    }
  })
])
[
  {
    _id: null,
    lowestMarks: 68
  }
]
studentDB>

```

```

> db.students.aggregate([
  {
    $group: {
      _id: "$department",
      averageMarks: { $avg: "$marks" }
    }
  }
])
< {
  _id: 'ECE',
  averageMarks: 74
}
{
  _id: 'CSE',
  averageMarks: 91.4
}
{
  _id: 'IT',
  averageMarks: 84.66666666666667
}
{
  _id: 'AI&DS',
  averageMarks: 90
}
StudentDB>

```

```

> db.students.aggregate([
  {
    $group: {
      _id: null,
      highestMarks: { $max: "$marks" }
    }
  }
])
< {
  _id: null,
  highestMarks: 95
}
StudentDB> |

```

```

> db.students.aggregate([
  {
    $group: {
      _id: null,
      totalMarks: { $sum: "$marks" }
    }
  }
])
< {
  _id: null,
  totalMarks: 1113
}
StudentDB> |

```

SECTION 17

```
> db.students.aggregate([
  { $match: { department: "CSE" } },
  {
    $group: {
      _id: "$department",
      averageMarks: { $avg: "$marks" },
      totalStudents: { $sum: 1 }
    }
  },
  { $sort: { averageMarks: -1 } }
])
< {
  _id: 'CSE',
  averageMarks: 91.4,
  totalStudents: 5
}
StudentDB> |
```

```
> db.students.aggregate([
  { $sort: { marks: -1 } },
  { $limit: 3 },
  {
    $project: {
      _id: 0,
      rollNo: 1,
      name: 1,
      marks: 1
    }
  }
])
< {
  rollNo: 105,
  name: 'Kiran',
  marks: 95
}
{
  rollNo: 101,
  name: 'John',
  marks: 95
}
{
  rollNo: 110,
  name: 'Meena',
  marks: 93
}
StudentDB> d
```

```
> { $sort: { marks: -1 } }
< -1
StudentDB> |
```

```
> db.students.aggregate([
  { $sort: { marks: -1 } },
  { $limit: 3 },
  {
    $project: {
      _id: 0,
      rollNo: 1,
      name: 1,
      marks: 1
    }
  }
])
< {
  rollNo: 105,
  name: 'Kiran',
  marks: 95
}
{
  rollNo: 101,
  name: 'John',
  marks: 95
}
{
  rollNo: 110,
  name: 'Meena',
  marks: 93
}
StudentDB> |
```

```
> { $limit: 3 }
< 3
StudentDB> |
```

SECTION 18

```
> show dbs
< StudentDB 56.00 KiB
  admin      40.00 KiB
  config     108.00 KiB
  j          16.00 KiB
  local      72.00 KiB
StudentDB>
```

```
> show collections
< students
> db
< StudentDB
> db.students.find().pretty()
< {
  _id: ObjectId('6a9538c497af2a8638092557'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 95,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a8638092558'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  status: 'Active',
  location: 'Vijayawada'
}
{
  _id: ObjectId('6a9538c497af2a8638092559'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a863809255a'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  location: 'Warangal'
}
}
```

```
> db.students.countDocuments()
< 13
> db.students.distinct("department")
< [ 'AI&DS', 'CSE', 'ECE', 'IT' ]
> db.students.getIndexes()
< [ { v: 2, key: { _id: 1 }, name: '_id_' } ]
StudentDB>
```

SECTION 19

```
> db.students.insertOne({
  rollNo: 201,
  name: "Test Student",
  age: 20,
  department: "CSE",
  marks: 85
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a95410d97af2a8638092564')
}
StudentDB>
```

```
> db.students.find({ rollNo: 201 })
< {
  _id: ObjectId('6a95410d97af2a8638092564'),
  rollNo: 201,
  name: 'Test Student',
  age: 20,
  department: 'CSE',
  marks: 85
}
StudentDB>
```

```
> db.students.updateOne(
  { rollNo: 201 },
  { $set: { marks: 95 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
StudentDB>
```

```
> db.students.find({ rollNo: 201 })
< {
  _id: ObjectId('6a95410d97af2a8638092564'),
  rollNo: 201,
  name: 'Test Student',
  age: 20,
  department: 'CSE',
  marks: 95
}
StudentDB>
```

```
> db.students.deleteOne({ rollNo: 201 })
< {
  acknowledged: true,
  deletedCount: 1
}
StudentDB>
```

```
> db.students.find({ rollNo: 201 })
<
StudentDB>
```

SECTION 20

Category	Command	Purpose
Database	<code>use studentDB</code>	Select/create database
Collection	<code>db.createCollection()</code>	Create collection
Create	<code>insertOne()</code>	Insert one document
Create	<code>insertMany()</code>	Insert multiple documents
Read	<code>find()</code>	Retrieve documents
Read	<code>findOne()</code>	Retrieve one document
Update	<code>updateOne()</code>	Update one document
Update	<code>updateMany()</code>	Update multiple documents
Delete	<code>deleteOne()</code>	Delete one document
Delete	<code>deleteMany()</code>	Delete multiple/all documents

Sort	<code>sort()</code>	Sort results
Limit	<code>limit()</code>	Limit results
Skip	<code>skip()</code>	Skip results
Projection	<code>find({}, {...})</code>	Select/hide fields
Index	<code>createIndex()</code>	Create index
Aggregation	<code>aggregate()</code>	Process/group documents

SECTION 21

```
StudentDB
show collections
students
db.students.countDocuments()
13
db.students.find()
{
  _id: ObjectId('6a9538c497af2a8638092557'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 95,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a8638092558'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  status: 'Active',
  location: 'Vijayawada'
}
{
  _id: ObjectId('6a9538c497af2a8638092559'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  location: 'Hyderabad'
}
```

```
}
db.students.find({ department: "CSE" })
{
  _id: ObjectId('6a9538c497af2a8638092557'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 95,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a8638092558'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  status: 'Active',
  location: 'Vijayawada'
}
{
  _id: ObjectId('6a9538c497af2a863809255b'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a863809255e'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a8638092561'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  status: 'Active',
  location: 'Hyderabad'
}
```

```
> db.students.find({ marks: { $gt: 85 } })
< [
  {
    _id: ObjectId('6a9538c497af2a8638092557'),
    rollNo: 101,
    name: 'John',
    age: 20,
    department: 'CSE',
    marks: 95,
    status: 'Active',
    location: 'Hyderabad'
  }
]
{
  _id: ObjectId('6a9538c497af2a8638092558'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  status: 'Active',
  location: 'Vijayawada'
}
{
  _id: ObjectId('6a9538c497af2a863809255a'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  location: 'Warangal'
}
{
  _id: ObjectId('6a9538c497af2a863809255b'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a863809255e'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  status: 'Active',
  location: 'Hyderabad'
}
```

SECTION 22

MongoDB is **case-sensitive** for database and collection names.

Use `$set` to add or update fields.

Use `$inc` to increase or decrease numeric values.

Use `$rename` to rename fields.

Use `$unset` to remove fields.

Use `$gt`, `$gte`, `$lt`, `$lte`, `$ne`, and `$eq` for comparisons.

Use `$and`, `$or`, `$not`, and `$nor` for logical conditions.

Use `$in` and `$nin` for matching multiple values.

Use `sort()`, `limit()`, and `skip()` for controlling query results.

Use projection to select or hide fields.

Indexes improve query performance.

Aggregation is used for grouping, calculations, filtering, and multi-stage processing.

`deleteMany({})` deletes **all documents** in a collection, so use it carefully.

SECTION 23

CRUD Operations

```

> db.students.insertOne({
  rollNo: 201,
  name: "Test Student",
  age: 20,
  department: "CSE",
  marks: 85
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a9542cd97af2a8638092565')
}
StudentDB>

```

```

> db.students.find()
< {
  _id: ObjectId('6a9538c497af2a8638092557'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 95,
  status: 'Active',
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a8638092558'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  status: 'Active',
  location: 'Vijayawada'
}
{
  _id: ObjectId('6a9538c497af2a8638092559'),
  rollNo: 103,
  name: 'Bahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  location: 'Hyderabad'
}
{
  _id: ObjectId('6a9538c497af2a863809255a'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 80,
  location: 'Marangal'
}
{
  _id: ObjectId('6a9538c497af2a863809255b'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  status: 'Active',
  location: 'Hyderabad'
}

```

```

> db.students.updateOne(
  { rollNo: 101 },
  { $set: { marks: 95 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
StudentDB>

```

```

> db.students.updateMany(
  { department: "CSE" },
  { $set: { status: "Active" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 6,
  modifiedCount: 1,
  upsertedCount: 0
}
StudentDB>

```

```

> db.students.deleteOne({ rollNo: 103 })
< {
  acknowledged: true,
  deletedCount: 1
}
StudentDB>

```

```

> db.students.deleteMany({ marks: { $lt: 40 } })
{
  acknowledged: true,
  deletedCount: 0
}
StudentDB>

```